Reg. No. | | | | | | | | | | | [14EE82]

EIGHTH SEMESTER B. Tech EXAMINATION, JUNE - 2019

RENEWABLE ENERGY RESOURCES

Time: 3 Hours **Maximum Marks: 70**

Instructions:

1. Answer any **FIVE** full Questions choosing Atleast **TWO** questions from each part.
2. Missing data may be suitably assumed.

PART - A

1.	a.	Discuss the need for non-conventional energy sources.	04
	b.	Write a note about the availability of non-conventional energy sources in India.	04
	c.	Comment on the essentialities of energy storage. Mention the most significant energy storage.	06
2.	a.	Define the following solar terminologies: **i.** solar altitude angle (α) **ii.** solar azimuth angle (γ)	04
	b.	What is the significance of solar radiation measurement? Briefly explain the instrument used for measurement of beam radiation – Angstrom Pyrheliometer	06
	c.	Explain the day light hour (sunshine hours) in Srinagar on January 1 and July 1, where the latitude of Srinagar is 34° 05' N	04
3.	a.	Explain the principle of operation of solar cooker. List some of the merits of solar cooker.	08
	b.	What are the advantages of Green Houses? Explain how 'Green House' provides higher cultivation in comparison with traditional agriculture.	06
4.	a.	Explain electric power production scheme using solar pond with a schematic diagram.	08
	b.	With the help of a block diagram, explain the operation of stand-alone solar photovoltaic system. Mention its advantages.	06

PART - B

5.	a.	Show that the maximum power developed in a horizontal axis wind turbine is 0.593 times the total power available in the wind.	06
	b.	Mention the significant factors to be considered before the erection of a wind energy conversion system.	08
6.	a.	Briefly outline the different methods used for the extraction of energy from Biomass.	06
	b.	Explain the constructional details and working of KVIC digester plant.	08
7.	a.	Describe the closed cycle Ocean Thermal Energy Conversion (OTEC). What are the advantages of closed cycle over open cycle OTEC?	08
	b.	Mention the advantages and limitations of Tidal power plant.	06
8.	a.	With a suitable diagram, explain the electric power generated using Fuel Cells. Mention its advantages.	08
	b.	What is Magneto-Hydro Dynamic (MHD) power generation? Briefly explain the principle of MHD power generator.	06

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